Lab 5

Name: Abdulsamad Mousa Tuhaishi

ID: 2240001093

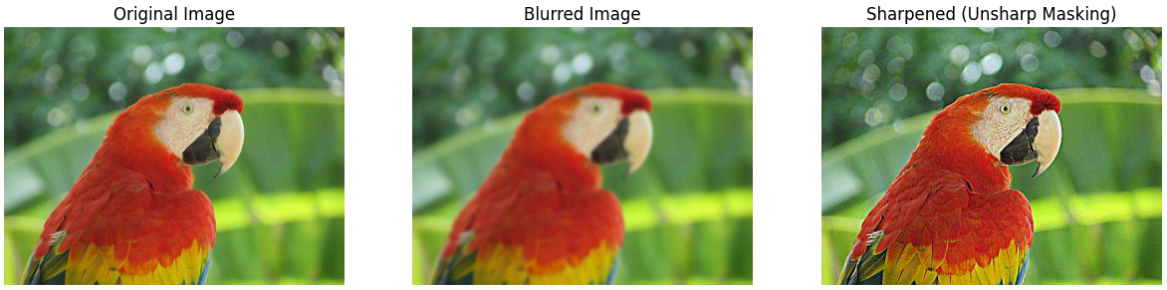
In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
# Procedural Task 1: Unsharp Masking
image = cv2.imread('images/Parrot.png')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

sigma = 2
amount = 1.5

image_float = image.astype(np.float32) / 255.0
blurred = cv2.GaussianBlur(image_float, (0, 0), sigmaX=sigma, sigmaY=sigm
sharpened = np.clip(image_float + amount * (image_float - blurred), 0, 1)

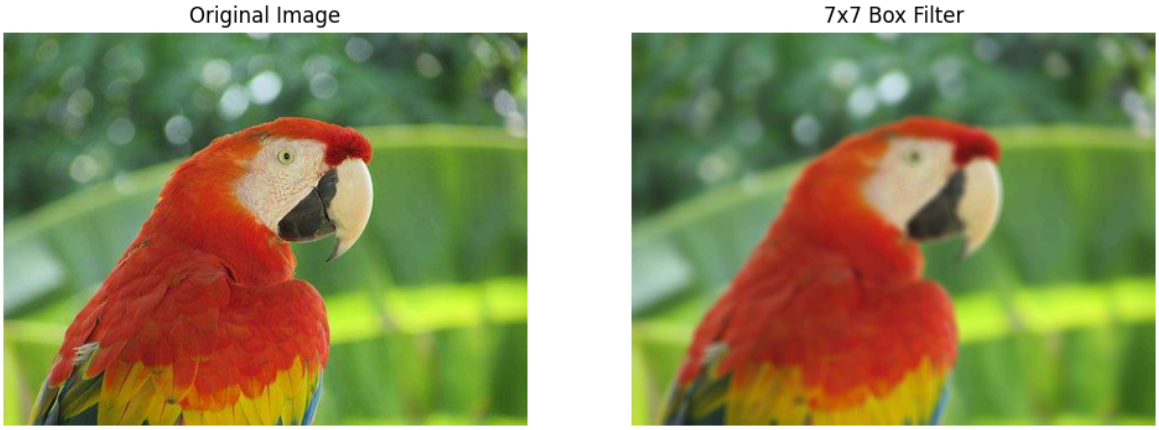
fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")
ax[1].imshow(blurred)
ax[1].set_title("Blurred Image")
ax[1].axis("off")
ax[2].imshow(sharpened)
ax[2].set_title("Sharpened (Unsharp Masking)")
ax[2].axis("off")
plt.show()
```



In [3]:
```
# Assessment Task 1: 7x7 box filter
kernel = np.ones((7, 7), np.float32) / 49
box = cv2.filter2D(image, -1, kernel)

fig, ax = plt.subplots(1, 2, figsize=(12, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")
ax[1].imshow(box)
ax[1].set_title("7x7 Box Filter")
```

```
ax[1].axis("off")
plt.show()
```



In [4]:
```
# Assessment Task 2: 5x5 and 21x21 Gaussian filters
gauss5 = cv2.GaussianBlur(image, (5, 5), 0)
gauss21 = cv2.GaussianBlur(image, (21, 21), 0)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")
ax[1].imshow(gauss5)
ax[1].set_title("Gaussian 5x5")
ax[1].axis("off")
ax[2].imshow(gauss21)
ax[2].set_title("Gaussian 21x21")
ax[2].axis("off")
plt.show()
```



In [5]:
```
# Assessment Task 3: 3x3 Laplacian sharpening
image = cv2.imread('images/Parrot.png')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
image_float = image.astype(np.float32) / 255.0

laplacian = cv2.Laplacian(image_float, cv2.CV_32F, ksize=1)
sharpened = np.clip(image_float - laplacian, 0, 1)

# scale the Laplacian to [0, 1] for display
lap_show = np.clip(0.5 + laplacian / (2 * np.percentile(np.abs(laplacian)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")
ax[1].imshow(lap_show)
ax[1].set_title("Laplacian Filter")
ax[1].axis("off")
```

```
ax[2].imshow(sharpened)
ax[2].set_title("Sharpened Image")
ax[2].axis("off")
plt.show()
```

